CSE230: Discrete Mathematics

Practice Sheet 7: **Number Theory**

Q1	Determine whether 4 | 7 and whether 4 | 24
Q2	What are the quotient and remainder when a) 193 is divided by 7? b) -42 is divided by 5? c) 597 is divided by 23? d) 30201 is divided by 33? e) 0 is divided by 18? f) 3 is divided by 8?
Q3	What are the possible remainders (according to the Division Algorithm) when an integer is a) Divided by 4? b) Divided by 9?
Q4	For each of the following, find the quotient and remainder (guaranteed by the Division Algorithm) a. When 17 is divided by 3. b. When -17 is divided by 3. c. When 73 is divided by 7. d. When -73 is divided by 7. e. When 436 is divided by 27. f. When 539 is divided by 110.
Q5	Today is Tuesday. Your uncle will come after 45 days. On which day (of the week) your uncle will be coming?
Q6	Find the value of $(177 \bmod 31 \cdot 270 \bmod 31) \bmod 31$
Q7	Show that if $a \mid b$ and $b \mid a$, where a and b are integers, then $a = b$ or $a = -b$.
Q8	Prove or disprove that if $a \mid bc$, where a, b and c are positive integers and $a \neq 0$, then $a \mid b$ or $a \mid c$.
Q9	Prove that if a is an integer that is not divisible by 3, then $(a + 1)(a + 2)$ is divisible by 3.
Q10	Let m be a positive integer. Show that $a \bmod m = b \bmod m$ if $a \equiv b \pmod{m}$.
Q11	Considering a and b are integers, and $a \equiv b + 7 \pmod{19}$, show that $(a^2 - 11) \equiv (b^2 - 5b) \pmod{19}$

Q12	If we consider $50 \equiv 23 \pmod 9$, then prove that $34 \equiv 16 \pmod 9$ [Hint: Use the Theorem 5]
Q13	28. Find a **div** m and a **mod** m when a) $a = -991$, $m = 99$. b) $a = -119$, $m = 101$. c) $a = 10299$, $m = 999$. d) $a = 12346$, $m = 101$
Q14	Find the integer a such that a) $a \equiv 17 \pmod{29}$ and $-14 \leq a \leq 14$. b) $a \equiv -11 \pmod{21}$ and $90 \leq a \leq 110$. c) $a \equiv 24 \pmod{31}$ and $-15 \leq a \leq 15$. d) $a \equiv 99 \pmod{41}$ and $100 \leq a \leq 140$.
Q15	Use modular exponentiation to find: i) 7^64 **mod** 645. ii) 32^203 **mod** 99. iii) 22^3219 **mod** 243.
Q16	i) What is the octal and hexadecimal expansion of $(11011000110\ 01011)_2$. ii) What is the hexadecimal expansion of $(125110)_{10}$. iii)What is the decimal expansion of $(7716)_8$. iv) What is the binary expansion of $(BADDAD)_{16}$. v) What is the octal expansion of $(12344321)_{10}$.
Q17	Expand the decimal number $(506070)_{10}$ to the base $\boldsymbol{x}$, where $\boldsymbol{x} = ((10*2^2)^2 + 71)$ **mod** 23.
Q18	If $(ab)_4$ is a base-4 integer and $(ba)_7$ is a base-7 integer, what is the largest possible value of $(a+b)_{10}$ and why? Find non-zero values for a and b such that $(ab)_4=(ba)_7$, or prove that there are none.
Q19	Determine whether the integers in each of these sets are **pairwise relatively prime**: i) 14, 17, 85. ii) 21, 34, 55. iii) 25, 41, 49, 64. iv) 17, 18, 19, 23.
Q20	Consider two positive integers 5271 and 48714. Calculate the Greatest Common Divisor for the integers using Euclidean Algorithm. Also show the Least Common Multiple.
Q21	Find the greatest common divisor of the following pair of numbers using the **Euclidean Algorithm**:

	i) 11111, 111111. ii) 1529, 14038. iii) 750, 900. iv) 414, 662.
Q22	How many divisions are required to find gcd(21, 34) using the **Euclidean algorithm**?
Q23	Find gcd(92928, 123552) and lcm(92928, 123552), and verify that gcd(92928, 123552) · lcm(92928, 123552) = 92928 · 123552
Q24	If the product of two integers is 2^2 · 3^5 · 5^2 · 7^1 · 23^2 · 29^1 and their greatest common divisor is 23345, what is their least common multiple?
Q25	Determine whether the integers in each of these sets are pairwise relatively prime. a) 11, 15, 19 b) 14, 15, 21 c) 12, 17, 31, 37 d) 7, 8, 9, 11
Q26	If p is a prime number, $p > 3$, then show that $p^2 \equiv p^4 \equiv 1 \pmod 3$
Q27	Work out the GCD of 7105, 3185 and 2898 **only using the Euclidean Algorithm**. Note that, gcd(a, b, c) = gcd(gcd(a, b), c)
Q28	Given that, the square root of a number $\boldsymbol{n}$ when divided by 11 gives a remainder of 6 where $\mathbf{6 < \sqrt{n} < 28}$. Find the number $\boldsymbol{n}$. Then, using modular exponentiation, calculate: $\boldsymbol{n}^{125} \ \boldsymbol{mod}\ \mathbf{27}$.